

ROCKFALLGUARD: PROACTIVE OPEN-PIT MINE GEOLOGICAL RISK WARNING PLATFORM

TECHNICAL SPECIFICATION & COMPREHENSIVE PROJECT SYSTEM REPORT

Project Version: 2.2.0 (Production Ready)	Deployment Stack: FastAPI, PyTorch, XGBoost, Redis 7, Docker
ML Accuracy Achieved: 99.50% (Holdout Test)	Failure Event Recall: 100.0% (Zero Missed Collapses)
Vision CV Engine: PyTorch PitWallCNN (4,617 Images)	Emergency Channels: SMTP Email & Twilio SMS Gateway

1. Executive Summary & Problem Statement

In open-pit mining operations, bench slope failures and rockfalls represent catastrophic hazards that endanger miner lives, destroy heavy haulage machinery, and disrupt mining production. Traditional monitoring systems rely on static, reactive thresholds (e.g. alarming only after slope displacement has already exceeded critical limits) or isolated single-sensor alerts.

RockfallGuard solves this by introducing a proactive, multi-sensor data fusion platform powered by Machine Learning (XGBoost + LightGBM Ensemble) and Deep Learning (PyTorch CNN). It ingests multivariate time-series data from extensometers, piezometers, geophones, and weather sensors, filters out heavy machinery vibration noise, and classifies geological risk into three probabilistic levels: **Safe (<35%)**, **Warning (35–65%)**, and **Critical (>65%)**.

2. System Architecture & Signal Processing

The system pipeline comprises multi-sensor telemetry ingestion, digital signal filtering, feature engineering, ML inference, and automated emergency notification dispatch.

Subsystem	Technology / Algorithm	Operational Function
Geotechnical Sensing	Extensometers, Piezometers	Measures slope creep velocity (mm/h), acceleration (mm/h²), and pore pressure (kPa).
Vibration Filtering	4th-Order Butterworth Filter	Attenuates high-amplitude haul truck noise (>15 Hz) from geophone microseismic RMS.
Weather Integration	Open-Meteo REST API	Fetches real-time localized rainfall (mm/h) and humidity by mine GPS coordinates.
Data Imbalance	SMOTE Oversampling	Synthesizes rare slope failure instances to eliminate model bias toward safe states.
Cache & Pub/Sub	Redis 7 (In-Memory Engine)	Caches weather telemetry (5-min TTL) and broadcasts instant Pub/Sub alert messages.
GIS Mapping	Leaflet.js + OpenStreetMap	Renders interactive map pins and satellite bounds for every registered mine site.

3. Machine Learning & PyTorch CNN Performance

Metric Parameter	Target Threshold	Achieved Performance	Validation Method

SHAP Root-Cause Interpretability Example:

- **+2.31 Impact:** Elevated Acoustic Fracture Acceleration Index (Geophone Signal Spike)
- **+1.41 Impact:** Accelerated Displacement Rate (>1.106 mm/h² Extensometer Creep)
- **+0.84 Impact:** Hydraulic Pore Water Pressure & Groundwater Saturation Interaction

4. Automated Emergency Email & SMS Dispatch Engine

- **HTML Emergency Email (SMTP):** Dispatches structured HTML warnings detailing mine name, GPS coordinates, hazard probability, mandatory evacuation directives, and top SHAP physical triggers to registered mine geologists (``contact_email``).
- **Emergency SMS / Phone (Twilio Gateway):** Dispatches text alerts urging immediate lower bench evacuation to registered phone numbers (``contact_phone``).
- **Local Environment Configuration:** Configured via ``backend/.env`` (SMTP credentials & Twilio SID/Tokens) with automated fallback to simulated dispatch logs.

5. Docker Containerization & CI/CD Pipeline

- **Dockerfile & Docker-Compose:** Containerized FastAPI application and Redis 7 stack (``docker-compose up --build``).
- **GitHub Actions CI/CD (``github/workflows/ci-cd.yml``):** Automated testing workflow executing PyTest test suite and enforcing the >95% ML model accuracy verification constraint on every commit.

6. REST API Endpoint Specification

HTTP Method	Endpoint Path	Description / Output